ITESO
Universidad Jesuita de Guadalajara

MDE2542A Analog Systems Design
Examination 1

Take-home exam | 100 points

Name: _____ Student ID: _____

Show your calculations and justify your choices. Include the sources and component data you use. Label results as calculated or simulated. For Problem 2, include your editable SPICE circuit and models.

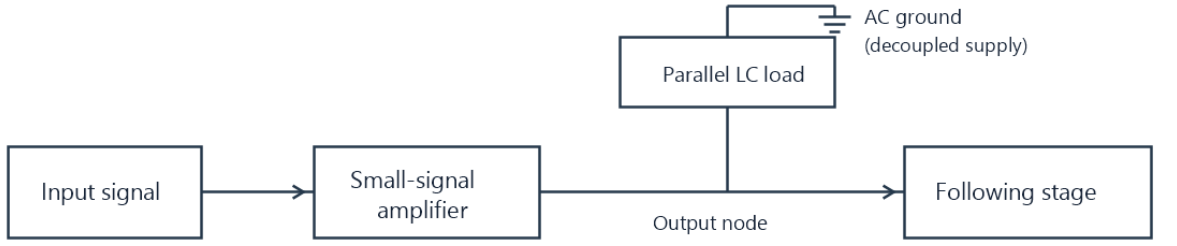
Problem 1 — The inductor will not arrive in time (40 points)

You are building a 5.8 GHz receiver. Its small-signal amplifier uses a parallel LC load, but the original inductor has a lead time of 52 weeks. You need another solution for the prototype.

Choose a new inductor and capacitor. Change both values and keep the parallel LC circuit. Use commercial SMD components.

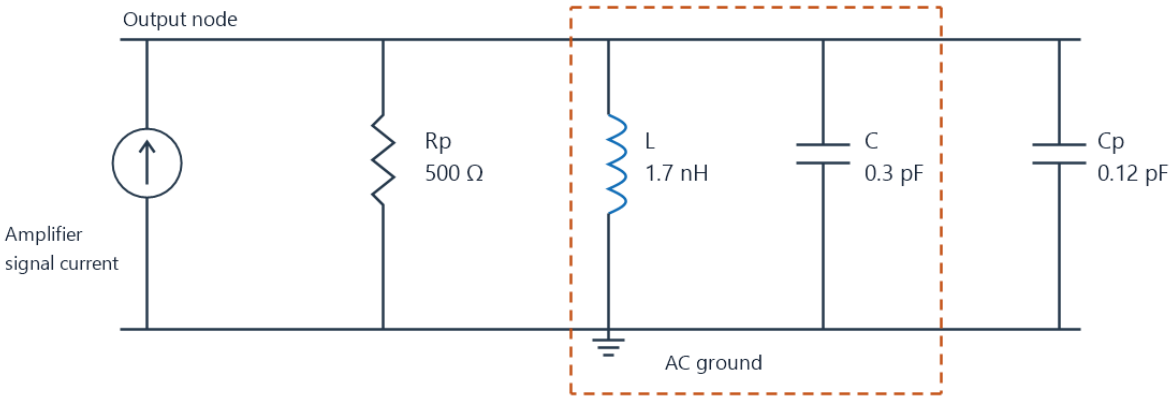
Where the LC load sits in the amplifier

Conceptual view — the amplifier and its bias circuit do not need to be designed.



Small-signal circuit used in this problem

All branches share the output node and AC ground. Values shown are for the original design.



Rp: amplifier output resistance. Cp: supplied PCB and amplifier parasitic capacitance.

The figure shows where the LC load connects to the amplifier output. Use the equivalent circuit below it for your calculations. All branches connect between the output node and AC ground. You do not need to design the amplifier or its bias circuit.

Parameter	Value
Target frequency	5.8 GHz
Allowed nominal resonance	5.626–5.974 GHz (±3%)
Original inductor	Coilcraft 0402DC-1N7XJRW, 1.7 nH, ±5%
Original capacitor	Murata GJM1555C1HR30WB01D, 0.3 pF, ±0.05 pF
Amplifier output resistance, Rp	500 Ω
PCB and amplifier parasitic capacitance, Cp	0.12 pF; keep it fixed
Conditions	Room temperature, small signal

The ±3% band applies to the nominal resonance. Report the range caused by component tolerances separately; that range does not have to stay within ±3%. The 52-week lead time is an assumption for this exercise.

A. Calculate the original design (8 points)

Calculate the ideal resonance, including Cp. Find the inductive and capacitive reactances at 5.8 GHz, and estimate the loaded Q with Rp = 500 Ω. What changes when you include component losses and internal parasitics?

B. Choose a new LC pair (8 points)

Choose new L and C values. Show your first calculation, then check the nominal resonance with a non-ideal model. It must stay within the stated band.

The peak impedance of your new tank must be at least 80% of the original. Use the same modelling method for both. Report their loaded Q and bandwidth, and explain the difference. You do not have to obtain the same bandwidth.

C. Validate commercial components (12 points)

Give the manufacturer and complete orderable part number for each replacement. Use manufacturer data to check tolerances, Q, self-resonance and suitability at 5.8 GHz. State the frequency of each Q value you use.

Include capacitor losses and internal parasitics. State where your model values came from and whether the models cover 5.8 GHz. Explain the difference between component Q and loaded circuit Q.

D. Additional considerations (tolerances and parasitics) (8 points)

For both designs, find the lowest and highest ideal resonance using the L and C tolerances, with Cp held fixed. Show which combinations give these limits.

Compare your ideal and non-ideal nominal results. If the prototype resonates at a different frequency, what would you check first?

E. Availability and cost (4 points)

Can you buy the two parts you selected? Include a quotation for each one:

Part number	Supplier	Stock / lead time	Unit price at 1,000 pieces	Cost for 1,000 circuits
Inductor				
Capacitor				
Total				

State the quotation date and currency. Each circuit uses one inductor and one capacitor. Use the price tier that applies when buying 1,000 of each part.

In two or three sentences, say whether you would order these parts for the prototype. Use your technical results and the availability to justify your answer. Identify any check you still need before ordering. Also state whether the quoted stock is enough for 1,000 circuits.

Hints

• Start with Ctotal = C + Cp and $f0 = 1/(2\pi\sqrt{(L\ Ctotal)})$.

• For ideal L and C, loaded $Q = Rp/(2\pi f0L)$.

• From a non-ideal impedance response, estimate Q = fpeak/BW using the two −3 dB frequencies.

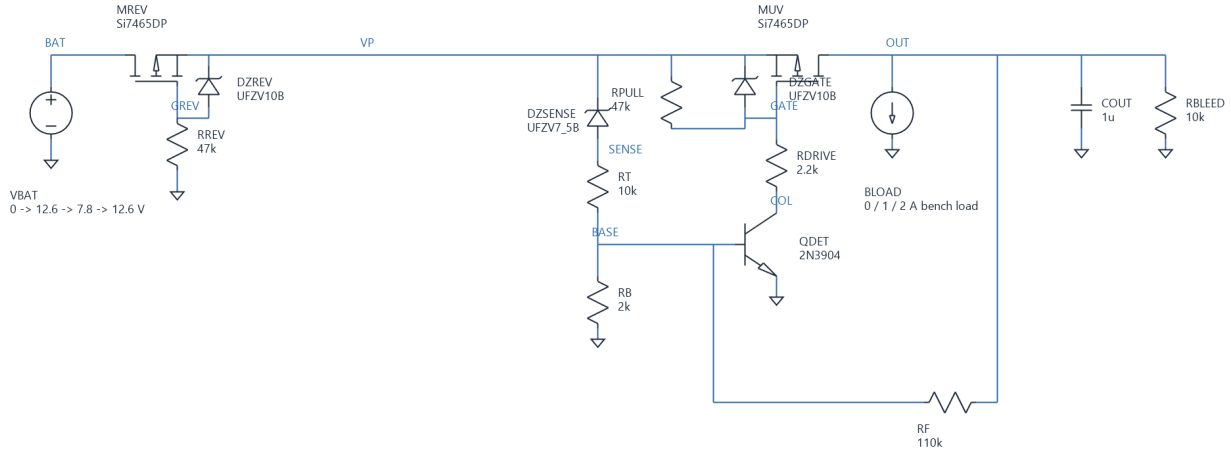
• Q at 2.4 GHz is not Q at 5.8 GHz. Look for an appropriate frequency curve or manufacturer model.

• Do not add internal parasitics again if the component model already includes them.

• You do not need S-parameters, impedance matching, Smith charts, transmission-line design or EM simulation.

Problem 2 — From a 3S battery to 6S (40 points)

The circuit below protects a board powered by a 3S LiPo battery. Now you need to use a 6S battery. Adapt the circuit and decide whether to use a diode or a MOSFET for reverse-polarity protection.



Use these conditions:

Parameter	3S reference	6S design
Maximum battery voltage	12.6 V	25.2 V
Nominal battery voltage	11.1 V	22.2 V
Disconnect when the battery falls to	9.9 V	19.8 V
Reconnect when the battery rises to	10.8 V	21.6 V
Load current	—	1 A nominal, 2 A continuous maximum
Ambient temperature	—	0–60 °C

Check the nominal thresholds at 27 °C, within ±5% of their targets. Measure them at the battery connector. Reconnect must remain above cutoff.

Assume balanced cells with their own protection. Use COUT = 1 µF and RBLEED = 10 kΩ. There is no other power source at the output. Start the reverse-battery test with COUT discharged. You do not need to study cable transients, hot-plugging or short circuits.

RF adds positive feedback. It helps the circuit switch cleanly and gives different voltages for cutoff and reconnection. This helps avoid repeated switching when the battery voltage recovers after the load disconnects. Keep RF when adapting the discrete circuit; its equations are given in Hints.

A. How does it work? (8 points)

Explain how the 3S circuit works. What do the PMOS devices, QDET and DZSENSE do? Show the current paths during normal operation, after UVLO cuts off, and with the battery reversed. Include the MOSFET body diodes.

B. Adapt it to 6S (10 points)

Calculate the new values and show your schematic. Which components need to change? Select commercial parts and check their voltage, current and power ratings. Include Zener bias, BJT drive and the gate-source voltage limits at a fully charged battery.

C. Diode or MOSFET? (14 points)

Compare both options for reverse-polarity protection:

- Output voltage and losses at 22.2 V input, with 1 A and 2 A loads.
- A simple temperature estimate at 60 °C ambient and 2 A. State the mounting conditions used for the thermal resistance. Allow for the MOSFET's higher resistance when hot.
- Relative battery runtime at the same useful load power. For this calculation, use a fixed 22.2 V battery, a 22.2 W external load and equal usable battery energy. Include the control circuit and bleeder losses.
- A complete BOM, including auxiliary parts. Quote prices for 1,000 pieces, in one currency. Give the cost per circuit and for 1,000 complete circuits. Include manufacturer, full MPN, supplier, date and stock/lead time. Exclude VAT, shipping and assembly.

Account for any shift in the battery-side UVLO thresholds caused by the protection's voltage drop.

Which option would you use? Justify your choice in no more than 100 words, using cost, heat and runtime.

You may choose an integrated solution if you demonstrate the same functions and limits. You must still explain the reference circuit and compare the diode and MOSFET options.

D. Validate your chosen design by simulation (SPICE)

8 points

Simulate the option you chose. Include your editable circuit and models. Show:

1. Cutoff and reconnection during a rising/falling battery sweep at 1 A.

2. Operation at 22.2 V and 2 A. Check device stresses at 25.2 V.

3. Startup at 18 V: the load stays disconnected. Report battery current.

4. A reversed battery of −25.2 V: report output voltage, leakage and device stresses. The load must not receive sustained forward power.

Use one summary table. Label results as calculated or simulated. You only need to simulate your chosen option.

Result	Diode	MOSFET
Cutoff / reconnect		
Result with −25.2 V		
VOUT at 22.2 V / 2 A		
Protection losses at 1 A / 2 A		
Temperature estimate and mounting assumption		
Relative runtime at 22.2 W		
Battery current at 18 V, after cutoff		
BOM cost per circuit / 1,000 circuits		

Hints

- You do not need to derive the hysteresis equations. For this series-Zener circuit, start with $V_{P,ON} \approx V_Z + V_{BE}(1 + R_T/R_B + R_T/R_F)$, and $V_{P,OFF} \approx V_{P,ON}/(1 + R_T/R_F)$. VP is after the input protection. These approximations neglect base current and switch drops.
- Use $V_{BE} \approx 0.60$ V for a first calculation. Check the Zener voltage at its actual current and verify the result with the component models.
- VGS(th) does not tell you the gate voltage needed for a low RDS(on).
- Diode loss: $P \approx V_F \cdot I$. MOSFET loss: $P \approx I^2 R_{DS(on)}$.
- For equal battery energy, $t_{MOS}/t_{DIODE} = P_{BAT,DIODE}/P_{BAT,MOS}$. Match delivered power, rather than load current. Report current after cutoff separately.
- Use the published price tier applicable to 1,000 pieces. If there is no exact 1,000-piece row, identify the applicable tier. Do not use a cheaper full-reel price unless it applies to that quantity.
- A simple thermal calculation is enough; no temperature sweep or PCB thermal simulation is required. You do not need to simulate a constant-power converter.
- COUT takes time to discharge after the switch turns off, especially without an external load.

Problem 3 — Will this LDO run too hot? (20 points)

The board of a portable MP3 player needs a 3.3 V supply at 400 mA from a 5 V input. The design uses an onsemi **NCP1117LPST33T3G** regulator in SOT-223.

The PCB layout is almost finished, and there is little space left around the regulator. Before sending the board for manufacturing, you are asked to check whether this choice can supply the required current continuously at the maximum ambient temperature. Would you let this design go into production?

Use the following conditions:

Parameter	Value
Input voltage	5.0 V
Output voltage	3.3 V
Continuous load current	400 mA
Maximum ambient temperature	60 °C
PCB	Two-layer FR-4, 1 oz copper
Copper available to spread heat	A single 10 mm × 10 mm square on the component side, connected to the regulator tab: 100 mm² in total
Cooling	Natural convection
Required margin	Junction temperature at least 10 °C below the upper operating limit

Evaluate steady-state operation at nominal voltages, with no other heat sources on the board. For the thermal estimate, use the datasheet's single-sided square-copper model as an approximation; do not add any benefit from the second layer or thermal vias.

A. Power dissipation (4 points)

Calculate how much power the regulator dissipates. Show the equation you use.

B. Thermal resistance (5 points)

Estimate the junction-to-ambient thermal resistance for the available copper area. Indicate which curve you used and include units. Is the regulator's current rating enough to decide whether it is suitable here? Explain briefly.

C. Junction temperature (5 points)

Calculate the junction temperature at the stated load and ambient temperature. Compare it with the operating limit and the required 10 °C margin. Explain why the absolute maximum temperature and the thermal shutdown temperature are different limits.

D. Your engineering decision (6 points)

Would you **APPROVE or REJECT** this design? Justify either answer with your numbers.

If you reject it, suggest one practical change and show how much improvement is needed. State whether your change keeps all the original requirements or changes one of them.

Would increasing the copper area to as much as 500 mm² be enough, keeping 1 oz copper and the same thermal model? Use the graph to support your answer; do not extrapolate beyond it.

Datasheet: onsemi NCP1117LP/D, Rev. 6, January 2024 (https://www.onsemi.com/download/data-sheet/pdf/ncp1117lp-d.pdf).

Hints

• Figure 32 in the datasheet includes thermal-resistance curves for different copper thicknesses.

• Although ground current is negligible, use 0.70 mA as an assumed value for this exercise.

• Assume that the input and output capacitors have already been selected correctly.

• You do not need to simulate or measure the PCB temperature.